Question **3**

Not yet answered

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Flag question

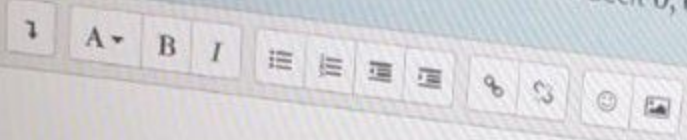
Sa se discute convergenta seriei:

$$\sum_{n\geq 1} \frac{q\cdot(q+p)\cdot(q+2p)\cdot(q+3p)\cdot\ldots\cdot(q+np)}{10\cdot 13\cdot 16\cdot 19\cdot\ldots\cdot(3n+10)\cdot n^3}$$

in functie de $p, q \in (0;\infty)$.

Fie $f:[0,1]\to\mathbb{R}$, $f(x)=\int_0^x \frac{e^{t^2}-1}{t^2}\,dt$, $\forall x\in[0,1]$.

Folosind eventual formula lui Taylor, determinati un polinom $P(x)$, astfel incit eroarea care se face aproximind $f(x)$ cu $P(x)$, pe intervalul $[0,1]$, sa fie mai mica decit $0,003$.



Question 1

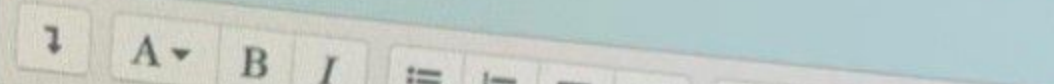
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Fie $f: [0,1] \to \mathbb{R}$, $f(x) = \int_0^x \frac{e^{t^2} - 1}{t^2} \, dt$, $\forall x \in [0,1]$.

Folosind eventual formula lui Taylor, determinati un polinom $P(x)$, astfel incit eroar
mai mica decit $0,003$.



Sa se studieze convergenta seriei:

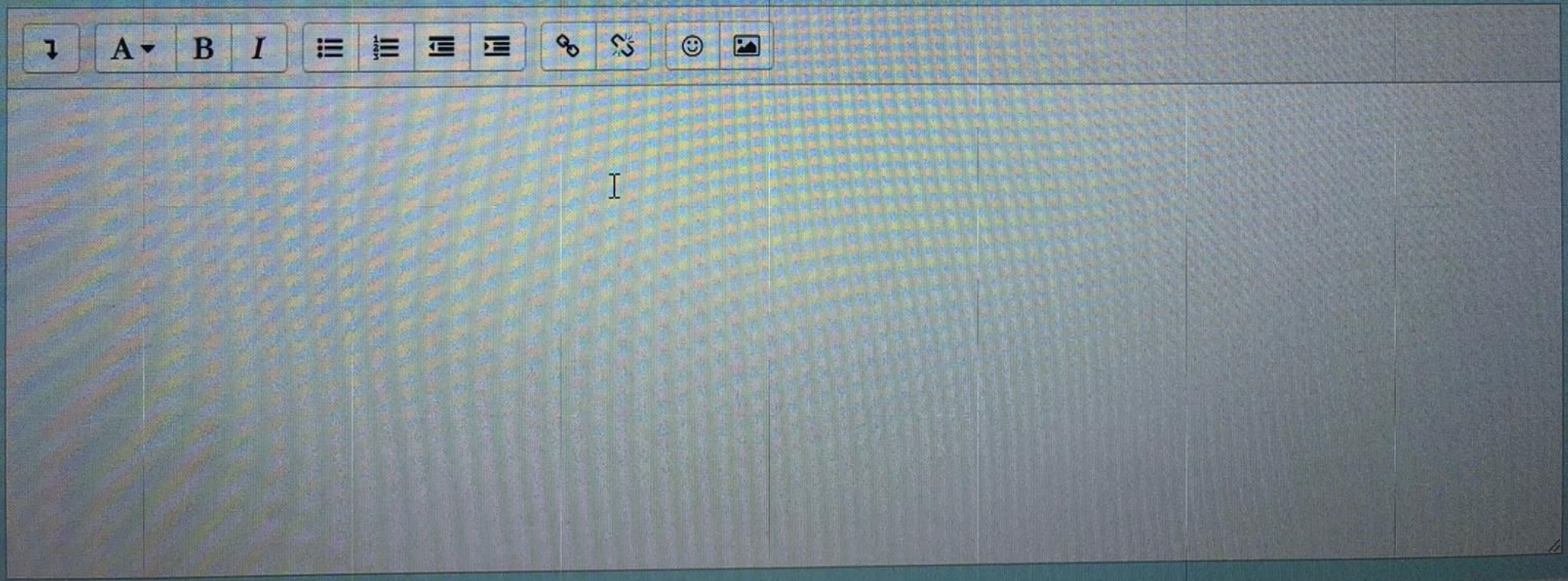
$$\sum_{n\geq 1} \frac{\sqrt{n^5+3n^3-1}+(-1)^n n^p - n^2\cdot\sqrt{n}}{n^p\cdot\sqrt[3]{n+5}}$$

$p \in \mathbb{R}$.

S-L-A1-S1: Analiză matematică (Seria CD - 2020)

Fie $f\colon [0,1] \to \mathbb{R}$, $f(x) = \int_0^x \frac{e^{t^4}-1}{t^4}\,\mathrm{d}t$, $\forall x \in [0,1]$.

Folosind eventual formula lui Taylor, determinati un polinom $P(x)$, astfel incit eroarea care se face aproximind $f(x)$ cu $P(x)$, pe intervalul $[0,1]$, sa fie mai mica decit $0,002$.



CS-L-A1-S1: Analiză matematică (Seria CD - 2020)

Fie $(a_n)_{n\geqslant 1}$ o progresie aritmetica de numere reale.

Fie $A=\left\{x\in\mathbb{R}\,\middle|\,\text{seria }\sum_{n\geqslant 1}\frac{a_n x^n}{(2n)!}\text{ este convergenta}\right\}$.

Pentru $x\in A$, fie $f(x)$ suma seriei $\sum_{n\geqslant 1}\frac{a_n x^n}{(2n)!}$.

$a)$ Pentru $x<0$, $x\in A$, sa se determine $f(x)$ in functie de primul termen si de ratia progresiei aritmetice $(a_n)_{n\geqslant 1}$.

$b)$ Daca $f'(0)=5$ si $f''(0)=1$, sa se determine progresia aritmetica $(a_n)_{n\geqslant 1}$.



Sa se discute convergenta seriei:

$$\sum_{n\geqslant 1} \frac{8\cdot 13\cdot 18\cdot 23\cdot 28\cdot \ldots \cdot (5n+3)\cdot n^2}{q\cdot (q+p)\cdot (q+2p)\cdot (q+3p)\cdot \ldots \cdot (q+(n-1)p)}$$

in functie de $p, q \in (0\,;\infty)$.

Fie $f:[0,1]\to\mathbb{R}$, $f(x)=\int_0^x \frac{e^{t^5}-1}{t^5}\,dt$, $\forall x\in[0,1]$.

Folosind eventual formula lui Taylor, determinati un polinom $P(x)$, astfel incit eroarea care se face aproximind $f(x)$ cu $P(x)$, pe intervalul $[0,1]$, sa fie mai mica decit $0,0012$.

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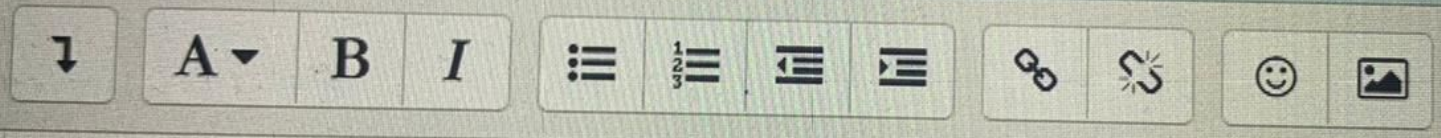
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Sa se discute convergenta seriei:

$$\sum_{n\geqslant 1} \frac{8\cdot 13\cdot 18\cdot 23\cdot 28\cdot \ldots \cdot (5n+3)\cdot n^2}{q\cdot (q+p)\cdot (q+2p)\cdot (q+3p)\cdot \ldots \cdot (q+(n-1)p)}$$

in functie de $p, q \in (0;\infty)$.



Question **2**

Not yet answered

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Sa se studieze convergenta seriei:

$$\sum_{n\geqslant 1} \frac{\sqrt[3]{n^2+2}+(-1)^n n^p-\sqrt[3]{n^2}}{n^p\cdot\sqrt{n+3}}$$

$p\in\mathbb{R}$.

Fie $(a_n)_{n\geqslant 1}$ o progresie aritmetica de numere reale.

Fie $A=\left\{x\in\mathbb{R}\,\middle|\,\text{seria }\sum_{n\geqslant 1}\frac{a_n x^n}{(2n)!}\text{ este convergenta}\right\}$.

Pentru $x\in A$, fie $f(x)$ suma seriei $\sum_{n\geqslant 1}\frac{a_n x^n}{(2n)!}$.

$a)$ Pentru $x<0$, $x\in A$, sa se determine $f(x)$ in functie de primul termen si de ratia progresiei aritmetice $(a_n)_{n\geqslant 1}$.

$b)$ Daca $f'(0)=5$ si $f''(0)=1$, sa se determine progresia aritmetica $(a_n)_{n\geqslant 1}$.

CS-L-A1-S1: Analiză matematică (Seria CD

Sa se studieze convergenta seriei:

$$\sum_{n\geq 1} \frac{\sqrt{n^5+3n^3-1}+(-1)^n n^p - n^2\cdot\sqrt{n}}{n^p\cdot\sqrt[3]{n+5}}$$

$p \in \mathbb{R}$.

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Sa se studieze convergenta seriei:

$$\sum_{n\geq 1} \frac{\sqrt{n^3+5n^2+1}+(-1)^{n-1}n^p-n\cdot\sqrt{n}}{n^p\cdot\sqrt[3]{n^2+3}}$$

$p \in \mathbb{R}$.

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Fie $f\colon [0,1] \to \mathbb{R}$, $f(x) = \int_0^x \frac{e^{t^3}-1}{t^3}\,dt$, $\forall x \in [0,1]$.

Folosind eventual formula lui Taylor, determinati un polinom $P(x)$, astfel incit ero
sa fie mai mica decit $0,002$.

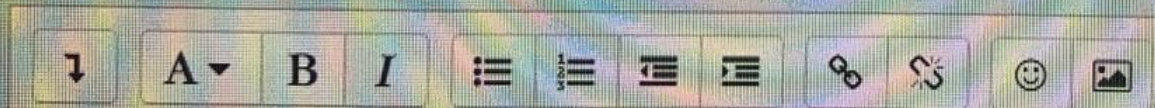
Fie $(a_n)_{n\geqslant 1}$ o progresie aritmetica de numere reale.

Fie $A=\left\{x\in\mathbb{R}\;\middle|\;\text{seria }\sum_{n\geqslant 1}\frac{a_n x^n}{(2n)!}\text{ este convergenta}\right\}$.

Pentru $x\in A$, fie $f(x)$ suma seriei $\sum_{n\geqslant 1}\frac{a_n x^n}{(2n)!}$.

$a)$ Pentru $x<0$, $x\in A$, sa se determine $f(x)$ in functie de primul termen si de ratia progresiei aritmetice $(a_n)_{n\geqslant 1}$.

$b)$ Daca $f'(0)=7$ si $f''(0)=\frac{3}{2}$, sa se determine progresia aritmetica $(a_n)_{n\geqslant 1}$.



Question **1**

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Sa se discute convergenta seriei:

$$\sum_{n\geqslant 1} \frac{9\cdot 13\cdot 17\cdot 21\cdot 25\cdot \ldots \cdot (4n+9)\cdot n^2}{p\cdot (p+q)\cdot (p+2q)\cdot (p+3q)\cdot \ldots \cdot (p+nq)}$$

in functie de $p, q \in (0;\infty)$.

CS-L-A1-S1: Analiză matematică (Seria CD -
)

Sa se discute convergenta seriei:

$$\sum_{n\geq 1} \frac{q\cdot(q+p)\cdot(q+2p)\cdot(q+3p)\cdot\ldots\cdot(q+np)}{10\cdot 13\cdot 16\cdot 19\cdot\ldots\cdot(3n+10)\cdot n^3}$$

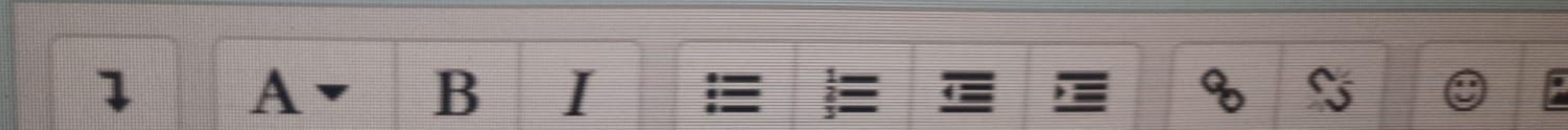
in functie de $p, q \in (0\,;\infty)$.

Previous Photo Next Photo

Sa se studieze convergenta seriei:

$$\sum_{n\geqslant 1} \frac{\sqrt{n^3+5n^2+1}+(-1)^{n-1}n^p-n\cdot\sqrt{n}}{n^p\cdot\sqrt[3]{n^2+3}}$$

$p\in\mathbb{R}$.



Sa se discute convergenta seriei:

$$\sum_{n\geqslant 1} \frac{p\cdot(p+q)\cdot(p+2q)\cdot(p+3q)\cdot\ldots\cdot\big(p+(n-1)q\big)}{11\cdot 13\cdot 15\cdot 17\cdot\ldots\cdot(2n+9)\cdot n^3}$$

in functie de $p, q \in (0\,;\infty)$.

Question **3**

Not yet answered

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Sa se discute convergenta seriei:

$$\sum_{n\geqslant 1} \frac{p\cdot(p+q)\cdot(p+2q)\cdot(p+3q)\cdot\ldots\cdot\big(p+(n-1)q\big)}{11\cdot 13\cdot 15\cdot 17\cdot\ldots\cdot(2n+9)\cdot n^3}$$

in functie de $p, q \in (0\,;\infty)$.

Fie $(a_n)_{n\geq 1}$ o progresie aritmetica de numere reale.

Fie $A = \left\{ x \in \mathbb{R} \,\middle|\, \text{seria } \sum_{n\geq 1} \frac{a_n x^n}{(2n+1)!} \text{ este convergenta} \right\}$.

Pentru $x \in A$, fie $f(x)$ suma seriei $\sum_{n\geq 1} \frac{a_n x^n}{(2n+1)!}$.

a) Pentru $x < 0, x \in A$, sa se determine $f(x)$ in functie de primul termen si de ratia progresiei aritmetice $(a_n)_{n\geq 1}$.

b) Daca $f'(0) = 1$ si $f''(0) = \frac{1}{6}$, sa se determine progresia aritmetica $(a_n)_{n\geq 1}$.

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